1. What is the primary engineering purpose of generating a sectional view of a component?

(A) To reveal internal features, cavities, and contours clearly without excessive hidden dashed lines
(B) To reduce drawing file size
(C) To show the 3D isometric perspective
(D) To determine material hardness

2. What is the imaginary plane that cuts through an object to expose its internal construction called?

(A) Cutting Plane (or Section Plane)
(B) Profile Plane
(C) Auxiliary Plane
(D) Miter Plane

3. According to BIS SP 46:2003, which line type is used to represent a Cutting Plane on an orthographic view?

(A) Chain thin line with thick ends and direction change (Type H)
(B) Continuous thick line (Type A)
(C) Dashed thin line (Type E)
(D) Continuous thin freehand line (Type C)

4. What do the arrows placed at the ends of a cutting plane line indicate?

(A) The viewing direction of the observer after the imaginary cut portion is removed
(B) The direction of molten metal flow
(C) The rotational axis of the machine
(D) The direction of gravity

5. In technical drawing, what are the parallel angled lines drawn on a cut surface called?

(A) Section lines (or Hatching lines)
(B) Extension lines
(C) Phantom lines
(D) Break lines

6. Which line type is used for drawing section lining (hatching) as per BIS standards?

(A) Continuous thin line (Type B, ~0.25 mm)
(B) Continuous thick line (Type A)
(C) Dashed thin line
(D) Chain thick line

7. What is the standard angle of inclination for hatching lines relative to the principal centerline or base outline?

(A) 45 degrees
(B) 30 degrees
(C) 60 degrees
(D) 90 degrees

8. What is the recommended spacing between adjacent hatching lines on standard mechanical drawings?

(A) 1.5 mm to 3 mm (depending on the size of the hatched area)
(B) 5 mm to 10 mm
(C) 0.2 mm to 0.5 mm
(D) 12 mm to 15 mm

9. When a cutting plane passes entirely across an object from one side to the other, the resulting view is a:

(A) Full Sectional View
(B) Half Sectional View
(C) Offset Sectional View
(D) Broken-out Sectional View

10. Which type of sectional view is specifically used for symmetrical objects to show both external and internal features in one view?

(A) Half Sectional View
(B) Full Sectional View
(C) Removed Sectional View
(D) Revolved Sectional View

11. In a half sectional view, what fraction of the object is imagined to be cut away and removed?

(A) One-quarter (1/4) of the object
(B) One-half (1/2) of the object
(C) Three-quarters (3/4)
(D) The entire object

12. In a half sectional view, which line type separates the unsectioned external half from the sectioned internal half?

(A) A centerline (Chain thin line)
(B) A continuous thick line
(C) A dashed hidden line
(D) A zigzag break line

13. Are hidden dashed lines normally drawn in the sectioned portion of a sectional view?

(A) No, hidden lines are omitted unless strictly necessary for dimensioning or clarity
(B) Yes, all hidden lines must be retained
(C) Only hidden circles are drawn
(D) They are drawn in red ink

14. When a cutting plane is bent or stepped at right angles to pass through non-collinear features, the section is called an:

(A) Offset Sectional View
(B) Full Section
(C) Revolved Section
(D) Broken-out Section

15. In an offset sectional view, how are the 90-degree stepped bends of the cutting plane represented in the sectional view?

(A) The stepped bends are NOT shown; the cut surface is depicted as continuous and seamless
(B) Drawn as thick solid dividing lines
(C) Drawn as dashed hidden lines
(D) Drawn as zigzag break lines

16. A cross-section rotated 90 degrees in place directly upon the existing orthographic view is called a:

(A) Revolved Section
(B) Removed Section
(C) Offset Section
(D) Partial Section

17. A cross-section rotated 90 degrees and positioned outside the view or adjacent to it is called a:

(A) Removed Section
(B) Revolved Section
(C) Full Section
(D) Quarter Section

18. Which type of section is bounded by an irregular continuous thin freehand line (Type C) to reveal a small local interior detail?

(A) Broken-out (Partial / Local) Section
(B) Half Section
(C) Offset Section
(D) Full Section

19. Which of the following machine components is NEVER sectioned longitudinally when cut by a cutting plane passing along its axis?

(A) Solid shaft (spindle)
(B) Hollow cylinder
(C) Valve body casting
(D) Pipe flange

20. Which of the following standard fasteners is exempt from longitudinal sectioning?

(A) Hexagonal bolt and nut
(B) Flanged pipe casing
(C) Piston head
(D) Connecting rod big-end cap

21. Why are solid shafts, bolts, rivets, and keys not sectioned when cut longitudinally?

(A) Because they are solid components without internal features; sectioning them adds clutter without providing useful information
(B) Because they are made of steel
(C) Because standard CAD cannot hatch them
(D) Because they have circular cross-sections

22. When a thin structural rib or web is cut along its longitudinal plane parallel to its flat face, how is it represented?

(A) It is left unhatched (unsectioned) to avoid a false impression of solid thickness
(B) It is hatched with double density
(C) It is hatched with horizontal lines
(D) It is filled solid black

23. If a cutting plane cuts across a thin rib or web PERPENDICULAR to its flat face (transversely), how is it shown?

(A) It is hatched normally like any other solid cross-section
(B) It is left unhatched
(C) It is drawn dashed
(D) It is omitted

24. In an assembly drawing with two adjacent mating parts, how should the section lining (hatching) be arranged?

(A) Hatched in opposite directions (e.g., 45 degrees on one part and 135 degrees on the other)
(B) Hatched in the exact same direction and pitch
(C) One part left completely unhatched
(D) Both parts filled solid black

25. How are very thin cross-sections (such as gaskets, sheet metal, and packing rings < 2 mm thick) represented in section?

(A) Filled completely solid black, with a narrow white gap separating adjacent pieces
(B) Hatched with 45 degree lines spaced 0.1 mm apart
(C) Shown as dashed lines
(D) Shown as phantom lines

26. Which AutoCAD command is used to apply cross-hatch patterns or solid color fills to enclosed areas?

(A) HATCH (H)
(B) FILL
(C) PATTERN
(D) SHADE

27. Which standard ANSI hatch pattern consists of parallel continuous lines inclined at 45 degrees, representing cast iron?

(A) ANSI31
(B) ANSI32
(C) ANSI33
(D) ANSI38

28. Which ANSI hatch pattern uses pairs of parallel lines at 45 degrees, representing mild steel?

(A) ANSI32
(B) ANSI31
(C) ANSI34
(D) ANSI37

29. Which AutoCAD command is used to edit the pattern, scale, or angle of an existing hatch object?

(A) HATCHEDIT (HE)
(B) EDIT
(C) MODIFYHATCH
(D) CHPROP

30. What is an 'Associative Hatch' in AutoCAD?

(A) A hatch that automatically updates its boundary and area whenever the surrounding bounding geometry is moved or stretched
(B) A hatch linked to a spreadsheet
(C) A hatch that cannot be erased
(D) A 3D volumetric fill

31. Which AutoCAD command converts a closed 2D loop of lines or arcs into a 2D planar surface region for Boolean operations?

(A) REGION (REG)
(B) BOUNDARY
(C) POLYLINE
(D) UNION

32. Which AutoCAD command allows stretching or moving vertices of entities while keeping connected lines intact?

(A) STRETCH (S)
(B) EXTEND
(C) SCALE
(D) MOVE

33. Which selection method is MANDATORY when using the STRETCH command in AutoCAD?

(A) Crossing Window (dragged right-to-left, green box)
(B) Window Selection (dragged left-to-right, blue box)
(C) Clicking individual entities
(D) Fence selection

34. Which AutoCAD command explodes compound objects like blocks, polylines, hatches, and dimensions into constituent parts?

(A) EXPLODE (X)
(B) BREAK
(C) DISASSEMBLE
(D) UNGROUP

35. Which AutoCAD command creates parallel copies of lines, polylines, concentric circles, or offset curves at a specified distance?

(A) OFFSET (O)
(B) COPY
(C) ARRAY
(D) PARALLEL

36. Which AutoCAD command extends objects to meet boundary edges of other entities?

(A) EXTEND (EX)
(B) LENGTHEN
(C) STRETCH
(D) SCALE

37. Which AutoCAD command joins connected collinear lines, arcs, splines, or polylines into a single unified object?

(A) JOIN (J)
(B) MERGE
(C) UNION
(D) ATTACH

38. Which AutoCAD command breaks an object into two pieces between two picked points or at a single point?

(A) BREAK (BR)
(B) TRIM
(C) CUT
(D) SPLIT

39. In AutoCAD's HATCH command, what does the 'Pick internal point' option do?

(A) Uses a flood-fill boundary detection algorithm to find all surrounding entities forming a closed loop around the clicked point
(B) Requires manual clicking of every boundary vertex
(C) Selects the entire layer
(D) Selects only rectangles

40. What happens if a user tries to apply HATCH using 'Pick internal point' on a boundary that has a tiny gap (is not closed)?

(A) AutoCAD displays a warning highlighting the gap with red circles and fails to hatch unless the gap is within tolerance
(B) The entire drawing canvas is filled with hatch
(C) AutoCAD automatically closes the gap with a solid line
(D) The drawing file is deleted

41. What does the 'Gap Tolerance' (HPGAPTOL) system variable in AutoCAD control during hatching?

(A) Allows AutoCAD to ignore gaps up to the specified size in non-closed boundaries and still apply hatching successfully
(B) Sets the gap between hatch lines
(C) Controls printer margin gaps
(D) Sets text font kerning

42. In AutoCAD HATCH Island Detection, what does the 'Normal' style do?

(A) Hatches inward from the outer boundary, turns OFF hatching at the first internal island, and turns ON hatching at the next nested island
(B) Hatches only the outermost area
(C) Hatches through all internal islands ignoring them completely
(D) Draws islands as circles

43. In AutoCAD HATCH Island Detection, what does the 'Outer' style do?

(A) Hatches only the space between the outer boundary and the first internal island, leaving all interior islands completely unhatched
(B) Hatches everything
(C) Hatches alternating islands
(D) Deletes internal islands

44. In AutoCAD HATCH Island Detection, what does the 'Ignore' style do?

(A) Hatches the entire bounded area completely, ignoring all internal island boundaries
(B) Hatches only islands
(C) Leaves everything blank
(D) Throws an error

45. How can you specify a custom Hatch Origin in AutoCAD to ensure that hatch lines align neatly with a specific corner vertex?

(A) Click 'Set Origin' in the Hatch Creation Ribbon tab and snap to the desired vertex
(B) Move the entire drawing to (0,0)
(C) Rotate the UCS
(D) Change the layer color

46. What is the consequence of applying EXPLODE to a hatch object in AutoCAD?

(A) Converts the hatch into thousands of individual, disconnected line entities, greatly increasing file size and losing associativity
(B) Changes the hatch color
(C) Removes the hatch cleanly
(D) Converts hatch into a 3D block

47. When creating an offset sectional view, why is the cutting plane stepped at 90 degrees?

(A) To pass through important interior features that do not lie along a single straight line
(B) To reduce the cutting length
(C) Because machine tools can only cut at 90 degrees
(D) To match the drawing sheet border

48. In an offset section, where are the change-of-plane steps shown on the orthographic view containing the cutting plane line?

(A) Shown as thick line segments at each 90-degree corner of the Type H cutting plane line
(B) Shown as zigzag lines
(C) Shown as circles
(D) Omitted completely

49. When drawing a Revolved Section directly on a view, what line type is used for its boundary if the view is NOT broken?

(A) Continuous thin line (Type B)
(B) Continuous thick line (Type A)
(C) Dashed thin line
(D) Chain thin line

50. If the view is broken open to accommodate a Revolved Section between the broken ends, what line type is used for the section boundary?

(A) Continuous thick line (Type A)
(B) Continuous thin line
(C) Dashed thin line
(D) Phantom line

51. Which of the following machine parts should be hatched when cut by a cutting plane?

(A) Cast iron machine frame / housing wall
(B) Solid cylindrical roller
(C) Hardened steel dowel pin
(D) Drive shaft key

52. In an assembly drawing of a knuckle joint, which pin connects the fork end and eye end, and is it sectioned longitudinally?

(A) Knuckle pin (cylindrical pin) - it is NOT sectioned longitudinally
(B) Knuckle pin - it is fully hatched
(C) Cotter pin - it is cross-hatched
(D) Taper sleeve - it is removed

53. In an assembly drawing of a flange coupling, which component is kept unsectioned when the cutting plane passes along its axis?

(A) Connecting bolt and nut
(B) Flange hub
(C) Keyway slot
(D) Pipe bore

54. When a cutting plane passes through the spokes or arms of a wheel or pulley, how are the spokes represented in section?

(A) The spokes are shown unsectioned (unhatched) to avoid giving a false impression of a solid web/disk
(B) The spokes are hatched at 45 degrees
(C) The spokes are drawn dashed
(D) The spokes are blacked out

55. When an object has an odd number of features (e.g. 3 or 5 equally spaced holes on a flange), how is a sectional view constructed?

(A) Using an Aligned Section: the cutting plane bends to pass through one hole and is rotated into the primary section plane
(B) Using an Offset Section
(C) By drawing a full section and showing one hole hidden
(D) By omitting the odd hole

56. In an Aligned Section, what happens to the angled portion of the cutting plane?

(A) It is rotated about the axis of symmetry until it aligns with the primary cutting plane before projecting
(B) It is projected at its true skewed angle
(C) It is foreshortened
(D) It is projected as an ellipse

57. In AutoCAD, what does typing '@' immediately at the second point prompt of the BREAK command accomplish?

(A) Breaks the selected entity at the exact single picked point without removing any gap segment
(B) Deletes the entity
(C) Cancels the break
(D) Connects to the origin

58. Which command in AutoCAD performs a Boolean subtraction between two overlapping 2D Regions?

(A) SUBTRACT (SU)
(B) UNION (UNI)
(C) INTERSECT (IN)
(D) TRIM

59. Which command in AutoCAD combines multiple overlapping 2D Regions into a single unified perimeter Region?

(A) UNION (UNI)
(B) SUBTRACT
(C) JOIN
(D) GROUP

60. Which command retains ONLY the common overlapping area of two or more intersecting 2D Regions?

(A) INTERSECT (IN)
(B) UNION
(C) SUBTRACT
(D) BOUNDARY

61. In AutoCAD, if you need to create a closed boundary polyline from an interior point without creating a Region, which command is used?

(A) BOUNDARY (BO)
(B) REGION
(C) PLINE
(D) RECTANG

62. What is the function of the 'Inherit Properties' button in AutoCAD's Hatch Creation ribbon tab?

(A) Allows the user to click an existing hatch in the drawing to immediately copy its pattern, scale, and angle for new hatches
(B) Inherits layer color only
(C) Deletes existing hatch
(D) Scales hatch by 2x

63. In sectional views, what happens when dimension text must be placed inside a hatched area?

(A) The hatching lines are interrupted (broken) around the dimension numeral so that the number remains perfectly legible
(B) The number is written across the hatch lines
(C) The hatch is removed completely
(D) Dimensions are not allowed on sections

64. If the outline of an object happens to be inclined at 45 degrees to the horizontal, at what angle should the hatching lines be drawn?

(A) At 30 degrees or 60 degrees to the baseline (to prevent hatching lines from being parallel to the outline)
(B) At 45 degrees
(C) At 0 degrees
(D) At 90 degrees

65. What is the standard ISO convention when three adjacent components meet at a common corner in an assembly section?

(A) Vary the hatching angle (e.g. 45? and 135?) and vary the line spacing/pitch or offset the lines for the third component
(B) Leave the third component blank
(C) Solid fill all three
(D) Draw all three with identical lines

66. In AutoCAD, what does the 'Hatch Transparency' property control?

(A) The opacity/transparency level of the hatch pattern (from 0 = fully opaque to 90 = mostly transparent), allowing underlying lines to remain visible
(B) The line weight of the hatch
(C) The distance between hatch lines
(D) The layer visibility

67. Which command in AutoCAD opens the traditional classic dialog box for Hatch and Gradient settings?

(A) HATCH (with HPDLGMODE = 1) or CLASSICHATCH
(B) HATCHEDIT
(C) PROPERTIES
(D) BHATCH

68. In AutoCAD, what system variable sets the default hatch pattern name?

(A) HPNAME
(B) HPSCALE
(C) HPANG
(D) HPDLGMODE

69. In AutoCAD, what system variable sets the default angle for newly created hatches?

(A) HPANG
(B) HPNAME
(C) HPSCALE
(D) HPGAPTOL

70. In AutoCAD, what system variable sets the default scale factor for hatch patterns?

(A) HPSCALE
(B) HPNAME
(C) HPANG
(D) LTSCALE

71. What is a 'Removed Section' drawn at a larger scale than the parent view called?

(A) An enlarged detail section (or Detail View)
(B) A revolved section
(C) A broken-out section
(D) An auxiliary projection

72. When a cutting plane passes through the axis of a standard helical compression spring, how are the cut coils shown?

(A) The coils are shown with cross-section hatching of circular or square wire, and the remainder is shown in outline
(B) The spring is left completely solid
(C) The spring cannot be sectioned
(D) The coils are filled with water

73. When drawing a half-section of a symmetrical bushing with an oil hole, why are hidden lines omitted from the unsectioned half?

(A) Because the internal details are already fully revealed and dimensionable in the sectioned half; showing them adds useless clutter
(B) Because CAD cannot draw them
(C) Because it is illegal under ISO
(D) To make drawing easier to erase

74. In AutoCAD, which command allows converting open intersecting lines into a closed polyline boundary suitable for hatching?

(A) PEDIT -> Multiple -> Join (or JOIN command)
(B) EXPLODE
(C) BREAK
(D) SCALE

75. What is the effect of changing the UCS before executing the HATCH command with an angle of 0 degrees?

(A) The hatch lines will align at 0 degrees relative to the current UCS X-axis, allowing effortless hatching along inclined faces
(B) The hatch will fail
(C) The drawing is rotated on monitor
(D) The hatch scale is inverted

76. A hollow stepped cylinder has an outer diameter of 100 mm for a length of 50 mm, and an outer diameter of 70 mm for a length of 50 mm. A through hole of diameter 40 mm is bored along its entire 100 mm length. What is the total cross-sectional area of metal exposed in a Full Longitudinal Section?

(A) 4948.01 sq. mm (Outer area 100x50 + 70x50 = 8500 sq. mm; Hole area 40x100 = 4000 sq. mm; Net cross-sectional area = 8500 - 4000 = 4500 sq. mm)
(B) 4500.00 sq. mm
(C) 5500.00 sq. mm
(D) 6000.00 sq. mm

77. For the cylinder in Q76, what is the hatched area exposed in a Half Longitudinal Section?

(A) 2250.00 sq. mm (4500 / 2 = 2250 sq. mm)
(B) 4500.00 sq. mm
(C) 1125.00 sq. mm
(D) 3000.00 sq. mm

78. A symmetrical cast iron flange has an outer diameter of 160 mm, a hub of diameter 80 mm rising 30 mm above a 20 mm thick flange base, and a central bore of 50 mm diameter. In a Half Sectional Front View, what is the horizontal width of the hatched flange base on the right side of the centerline?

(A) 55.00 mm ((160 - 50) / 2 = 110 / 2 = 55.00 mm)
(B) 80.00 mm
(C) 40.00 mm
(D) 25.00 mm

79. For the flange in Q78, what is the horizontal width of the hatched hub section on the right side of the centerline?

(A) 15.00 mm ((80 - 50) / 2 = 30 / 2 = 15.00 mm)
(B) 40.00 mm
(C) 25.00 mm
(D) 30.00 mm

80. In AutoCAD, a closed Region is created from a 120 mm x 80 mm rectangle (Area = 9600 sq. mm), and two circular Regions of diameter 25 mm (Area = 2 * pi * 12.5^2 = 981.75 sq. mm) are SUBTRACTED from it. What is the remaining surface area reported by MASSPROP?

(A) 8618.25 sq. mm (9600 - 981.75 = 8618.25 sq. mm)
(B) 9600.00 sq. mm
(C) 7636.50 sq. mm
(D) 8125.50 sq. mm

81. When drafting a sectional view of an assembly containing a bronze bushing pressed into a cast iron housing with a steel shaft inside, what hatching patterns and angles should be assigned to the three components in AutoCAD?

(A) Housing: ANSI31 at 45?; Bushing: ANSI33 at 135?; Shaft: Unsectioned (or ANSI32 at 45? with different scale)
(B) All three with ANSI31 at 45?
(C) All three filled solid black
(D) Housing solid, bushing dashed, shaft clear

82. A casting has a base wall 25 mm thick. In an offset section, the cutting plane steps 30 mm along the X-axis before continuing. In the resulting sectional elevation, what is the appearance of the transition line between the two offset cut areas?

(A) No line is drawn; the hatching pattern continues seamlessly across both areas as if cut by a single flat plane
(B) A thick continuous line is drawn
(C) A zigzag line is drawn
(D) A dashed hidden line is drawn

83. In AutoCAD, when using the HATCH command with pattern ANSI31 (lines at 45?), what happens if the user sets the Hatch Angle property to 45 degrees in the ribbon?

(A) The resulting hatch lines are drawn at 90 degrees (vertical: 45? base pattern + 45? user angle = 90?)
(B) The lines remain at 45 degrees
(C) The lines become horizontal at 0 degrees
(D) The lines are drawn at 135 degrees

84. In AutoCAD, to create hatch lines inclined at 135 degrees using pattern ANSI31 (which is defined at 45 degrees), what Hatch Angle must be entered?

(A) 90 degrees (45? base + 90? = 135?)
(B) 135 degrees
(C) 45 degrees
(D) 180 degrees

85. A square pyramid (base 60 mm, height 80 mm) rests on its base on HP. A horizontal cutting plane cuts through the pyramid at a height of 40 mm above the base. What is the shape and size of the section exposed in the Top View?

(A) A regular square of side 30 mm with hatching lines at 45 degrees
(B) A regular square of side 60 mm
(C) A rectangle of 60x30 mm
(D) An octagon of side 15 mm

86. For the pyramid in Q85, what is the cross-sectional area of the hatched top view section?

(A) 900.00 sq. mm (30 mm x 30 mm = 900 sq. mm)
(B) 3600.00 sq. mm
(C) 1800.00 sq. mm
(D) 450.00 sq. mm

87. A right circular cone (base diameter 60 mm, height 70 mm) is cut by a section plane inclined to the axis at an angle greater than the semi-vertical angle, cutting through all generators. What geometric curve bounds the true shape of the section?

(A) An ellipse
(B) A parabola
(C) A hyperbola
(D) A circle

88. When the section plane cutting a right circular cone is parallel to one of the slant generators of the cone, what curve is formed on the section?

(A) A parabola
(B) An ellipse
(C) A hyperbola
(D) A circle

89. When the section plane cutting a right circular cone is parallel to the central axis of the cone, what geometric curve is formed?

(A) A hyperbola (or rectangular hyperbola if cone angle is 90?)
(B) An ellipse
(C) A parabola
(D) A cycloid

90. In AutoCAD, what is the command sequence to create a custom boundary around a complex cluster of intersecting lines for hatching without manual trimming?

(A) BOUNDARY -> Pick Points -> Click inside closed area -> Apply HATCH to newly created polyline
(B) HATCH -> All
(C) REGION -> All
(D) TRIM -> Everything

91. In AutoCAD, which option under the HATCH command ensures that the hatch automatically trims around text annotations placed inside the boundary?

(A) Text Island Detection (automatic boundary masking around MTEXT and DTEXT entities)
(B) WIPEOUT command
(C) EXPLODE text
(D) LAYER freeze

92. When a cylindrical pipe with an internal thread is shown in a longitudinal sectional view, how are the screw threads represented?

(A) Crest lines are continuous thick lines, root lines are continuous thin lines, and hatching extends all the way to the thick crest lines
(B) Hatching stops at the root lines
(C) Threads are shown dashed
(D) Hatching covers the entire bore

93. In an assembly drawing of a stuffing box, what material hatch pattern is traditionally used for asbestos or Teflon packing rings according to ISO standards?

(A) Cross-hatch pattern forming small squares / diamonds (or ANSI37 / SOLID)
(B) Parallel 45? single lines
(C) Double parallel lines
(D) Dots pattern

94. When an object has a thin cylindrical shell (e.g. thickness 1.2 mm), why is solid black filling used instead of angled hatch lines?

(A) Because the distance between lines would be too small to draw cleanly without blurring into an illegible ink blob
(B) Because it represents rubber
(C) Because ISO forbids hatching steel
(D) To save CAD memory

95. In AutoCAD, what command allows measuring the exact surface area and perimeter of a hatched Region or boundary?

(A) AREA (or MASSPROP)
(B) MEASURE
(C) DIST
(D) LENGTHEN

96. What is the consequence of applying the STRETCH command to an associative hatched boundary in AutoCAD?

(A) The boundary vertices move, and the hatch pattern automatically regenerates to fill the newly deformed boundary area seamlessly
(B) The hatch is deleted
(C) The hatch separates from the boundary
(D) AutoCAD crashes

97. When a cutting plane passes through a web or rib of a cast bracket transversely (perpendicular to its length), how is the rib represented?

(A) It is fully hatched with standard 45-degree section lining
(B) It is left unhatched
(C) It is shown dashed
(D) It is broken out

98. In AutoCAD, which command allows creating a wipeout polygon that masks or hides drawing entities underneath without deleting them?

(A) WIPEOUT
(B) MASK
(C) COVER
(D) HIDE

99. What happens if a user selects 'Separate Hatches' during the creation of hatch patterns across multiple distinct closed boundaries in AutoCAD?

(A) Creates individual, independent hatch objects in each boundary rather than a single compound multi-boundary hatch entity
(B) Deletes all but the first hatch
(C) Changes the pattern
(D) Merges all boundaries

100. Why must the dividing line in a half-sectional view NEVER be drawn as a solid continuous line?

(A) Because a solid line would indicate a physical edge or wall on the object, whereas the division is purely an imaginary boundary between two viewing representations
(B) Because ISO prohibits lines on sections
(C) Because CAD cannot print solid lines
(D) To distinguish it from the sheet border

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Explain the principle and necessity of sectional views in engineering drawings, detailing how section planes eliminate visual confusion caused by overlapping hidden lines.

Q2. Detail the representation and conventions of Cutting Plane lines according to BIS SP 46:2003 (ISO 128), explaining line thickness, dash-dot patterns, viewing direction arrows, and reference lettering.

Q3. Explain the rules and conventions for Section Lining (Hatching) according to BIS SP 46, detailing line types, angles of inclination, spacing, adjacent components, and thin section representations.

Q4. Compare and contrast the six primary types of sectional views: Full Section, Half Section, Offset Section, Revolved Section, Removed Section, and Broken-Out Section.

Q5. Detail the rules governing components and features that are EXEMPT from longitudinal sectioning (non-sectioned parts) according to BIS SP 46 and ISO 128, explaining the engineering rationale.

Q6. A cast iron flanged pipe joint assembly connects two pipes of 80 mm bore, 100 mm outer diameter, and 180 mm flange diameter (flange thickness 25 mm). The flanges are connected by four 16 mm bolts on a PCD of 140 mm with a 3 mm gasket between flanges. Explain how this assembly is represented in

Q7. Explain the Half Sectional View in exhaustive detail, including its construction methodology for symmetrical mechanical bodies, rules for dividing lines, and dimensioning protocols.

Q8. Explain the Offset Sectional View, detailing how the cutting plane path is defined in parent views, how 90-degree transitions are drafted, and how to represent non-collinear features.

Q9. Explain Revolved Sections and Removed Sections with clear geometric illustrations, comparing their drafting rules, line types, and practical workshop applications.

Q10. Detail the HATCH command in AutoCAD, explaining the internal flood-fill boundary detection algorithm, the Select Objects mode, and island detection styles (Normal, Outer, Ignore).

Q11. Detail the standard Hatch Patterns in AutoCAD (ANSI, ISO, Solid, Gradient), explaining their naming conventions, predefined scales, and typical material associations.

Q12. Explain the HATCHEDIT command in AutoCAD, detailing how to modify existing hatch patterns, re-associate detached boundaries, adjust pattern scales, and set custom origins.

Q13. Explain the REGION command in AutoCAD and its application in 2D mechanical section analysis, detailing Boolean operations (UNION, SUBTRACT, INTERSECT) and MASSPROP.

Q14. Detail the BREAK and JOIN commands in AutoCAD, explaining their operational syntax and specific applications in generating partial and broken-out sectional views.

Q15. Explain the Aligned Sectional View, detailing its construction for circular flanges with an odd number of holes or radial arms and contrasting it with standard offset sections.

Q16. A cast steel bracket has a horizontal base plate 140 mm x 80 mm x 20 mm thick, a vertical back plate 80 mm wide x 20 mm thick x 70 mm high, and a central triangular reinforcing web 12 mm thick connecting them. A 30 mm hole is bored through the base, and a 25 mm hole through the back plate.

Q17. Explain the drafting procedure in AutoCAD to create an Offset Sectional View of a machine housing having three non-collinear bored holes, detailing cutting plane layout and hatch application.

Q18. Discuss the rules for dimensioning sectional views in engineering drawings, detailing how to avoid placing dimension lines across hatched areas and how to handle leader callouts.

Q19. Explain the representation of fasteners (bolts, studs, screws, pins, keys) in sectional assembly drawings, explaining when they are shown in full elevation and when they are hatched.

Q20. Describe the Broken-Out (Partial) Sectional View, detailing its application for inspecting localized interior features, the convention for break lines, and step-by-step CAD execution.

Q21. Explain the methods of sectioning thin-walled structures (ribs, webs, gussets) cut transversely versus longitudinally, illustrating with a cast iron T-bracket.

Q22. Explain the representation of gears, bearings, and springs in sectional drawings, detailing conventional simplifications adopted in engineering practice.

Q23. Detail the command sequence and drafting steps in AutoCAD to create a Half-Sectional Front View of a stepped flange bushing (Bore 40 mm, Hub 70 mm, Flange 120 mm).

Q24. Explain the Hatch Boundary Leak detection tools in AutoCAD, discussing Gap Tolerance (HPGAPTOL), red circle glyphs, and methods to resolve non-closed loops.

Q25. Explain the methods of creating custom hatch patterns in AutoCAD, detailing the structure of the .pat file, delta-x, delta-y, and line dash-gap specifications.

Q26. A cast iron valve handwheel of 200 mm diameter has four elliptical spokes connecting a 40 mm diameter hub to a 20 mm circular rim. Explain how the spokes and rim are represented in a Full Sectional Elevation.

Q27. Explain the procedure to create an associative hatch with background color masking in AutoCAD, detailing the HATCHTOBACK command and draw order management.

Q28. Detail the STRETCH command application in sectional views, explaining how to modify wall thicknesses and hole locations while preserving hatch associativity.

Q29. Explain the methods of creating and sectioning 2D assembly views containing standard machine elements (bearings, seals, shafts) in AutoCAD.

Q30. Summarize the complete quality assurance audit checklist for validating sectional views prior to engineering manufacturing release.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[B]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

'C') are placed adjacent to the arrowheads, readable in the normal sheet viewing orientation.
- Section Title: The resulting sectional view is explicitly titled directly above or below the view: 'SECTION A-A' or 'SECTION B-B'.
- If multiple cutting planes exist on a single drawing, letters are assigned sequentially (A-A, B-B, C-C, avoiding confusing letters like I, O, Q, X).

- One half of the view displays internal features in section; the other half displays external visible contours.
- The two halves are separated by a central CHAIN THIN CENTERLINE (never a solid line).

3. Offset Sectional View:
- The cutting plane is stepped or bent at 90-degree right angles to pass through non-collinear holes, slots, or cavities.
- The 90? steps are NOT shown in the sectional view; the cut appears as a continuous flat plane.

4. Revolved (Rotated) Section:
- A cross-section of a structural feature (rib, spoke, arm, bar) is cut by a plane perpendicular to its axis, rotated 90 degrees in place, and drawn directly on the existing view.
- Drawn with continuous thin lines (Type B) without breaking the view.

5. Removed Section:
- A cross-section cut, rotated 90 degrees, and placed OUTSIDE the existing view (often adjacent or on an extension of the cutting plane line) and labeled 'SECTION A-A'.
- Drawn with standard continuous thick outlines (Type A). Can be drawn at an enlarged scale.

6. Broken-Out (Partial / Local) Section:
- Used when only a small localized interior detail needs inspection (e.g. a keyway, set-screw hole).
- A small section is revealed by an imaginary break line drawn as a continuous thin freehand wavy line (Type C).

>>**Solution to Question 5:** Detail the rules governing components and features that are EXEMPT from longitudinal sectioning (non-sectioned parts) according to B

4. Exception (Transverse Cutting Planes):
If the cutting plane passes TRANSVERSELY (crosswise / perpendicular to the axis) through a shaft, bolt, key, or rib, the exposed cross-section MUST BE HATCHED normally.

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2. Standard Construction Rules:
- Rule 1 (The Dividing Line):
* The line separating the unsectioned half from the sectioned half MUST BE DRAWN AS A CENTERLINE (Chain Thin Line, Type G)!
* Under no circumstances should a solid continuous line be drawn at this boundary; a solid line represents a physical edge, whereas this division is purely an imaginary boundary between two viewing representations.
- Rule 2 (Omission of Hidden Lines):
* Hidden dashed lines are STRICTLY OMITTED from both the sectioned half (where features are already visible outlines) and the unsectioned half (because the features are clearly defined in the adjacent sectioned half).
* Exception: A hidden line may be retained in the unsectioned half ONLY if needed to dimension an external feature that has no sectioned counterpart.

3. Dimensioning Protocol in Half Sections:
- For internal cylindrical features (bores, counterbores, keyways):
* Dimension lines are drawn originating from the sectioned feature, crossing slightly past the centerline, terminated with a SINGLE ARROWHEAD touching the sectioned wall!
* The dimension value is prefixed with the diameter symbol 'phi' (e.g. 'phi 50') to signify that the full diameter is represented, even though only half the bore is sectioned.

>>Solution to Question 8: Explain Revolved Sections. Describe how they differ from standard orthographic views, their specific uses, and the rules for drafting them.

[Detailed Step-by-Step Answer]:
1. Revolved (Rotated) Sections:
- Concept: A cutting plane is passed perpendicular to the longitudinal axis of an elongated feature (such as a crane hook, spoke of a wheel, structural rib, or hand tool shank).
- The resulting cross-section is rotated 90 degrees IN PLACE and drawn directly superimposed on the existing view.
- Drafting Conventions:
* If the view of the object is continuous (not broken), the outline of the revolved section is drawn with CONTINUOUS THIN LINES (Type B) and hatched. The original object outlines continue uninterrupted underneath.
* If the view of the object is broken open to receive the section, the cross-section is drawn between the break lines using CONTINUOUS THICK LINES (Type A).
- Application: Immediate inspection of cross-sectional profiles of wrenches, connecting rod shanks, and machine tool handles.

2. Removed Sections:
- Concept: The cross-section is cut by a perpendicular plane, rotated 90 degrees, and REMOVED from the view to be drawn elsewhere on the sheet.
- Placement Options:
* Placed adjacent to the cutting plane, connected to it by a chain thin centerline.
* Placed in a convenient empty space on the sheet, explicitly labeled with a title: 'SECTION A-A' or 'SECTION B-B'.
* Drawn at an ENLARGED SCALE (e.g., Scale 5:1) to show minute groove radii or thread details clearly (Detail Section).
- Drafting Conventions:
* Drawn with standard CONTINUOUS THICK OUTLINES (Type A) and fully hatched.
- Application: Complex turbine blades, structural steel I-beams, ship hull frames, and automotive chassis rails.

>>**Solution to Question 18:** Explain the HATCH command in AutoCAD, explaining the internal flood-fill boundary detection algorithm, the Select Objects mode, and is

inside a plate):
* Normal Island Detection: Hatches inward from outer boundary; stops hatching at 1st internal boundary; resumes hatching at 2nd nested boundary (alternating ring pattern).
* Outer Island Detection: Hatches only the outer space between the primary boundary and the 1st internal boundary; all internal islands are left completely clear. (Most common for mechanical sections with through-holes).
* Ignore Island Detection: Ignores all internal boundaries completely, hatching across the entire outer area as if no holes existed.

- Pattern Type & Name: Switch instantly between ANSI31, ANSI32, SOLID, or ISO patterns.
- Pattern Angle: Rotates the hatch pattern relative to active coordinate axes (e.g. changing 0? to 90? rotates ANSI31 from 45? to 135?).
- Pattern Scale: Adjusts the spacing density of hatch lines. Increasing scale widens line spacing; decreasing scale condenses spacing.
- Layer Assignment: Move hatch to designated 'HATCH' layer with assigned color and 0.25 mm lineweight.

3. Re-Associating Detached Boundaries:
- If an associative hatch loses its boundary link (e.g. if the original boundary was deleted or exploded), the hatch becomes non-associative.
- In HATCHEDIT, clicking 'Recreate Boundary' automatically generates a new closed polyline or region perfectly matching the hatch perimeter!
- Clicking 'Associative' re-links the hatch to this new boundary.

4. Custom Hatch Origin:
- In HATCHEDIT, clicking 'Hatch Origin -> Specified origin' allows clicking an exact vertex on the object.
- Hatch lines will originate precisely from that vertex, ensuring neat, symmetrical hatching layouts across multiple view panels.

>>**Solution to Question 13:** Explain the REGION command in AutoCAD and its application in 2D mechanical section analysis, detailing Boolean operations (UNION

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3. Key Drafting Rules:
- The rotated hole or spoke is drawn at its TRUE RADIAL DISTANCE from the center axis.
- No foreshortening is applied to the rotated feature.
- The resulting view is symmetrical, balanced, and displays the true radial geometry of all critical features clearly.

- Switch to layer 'CUTTING_PLANE' (Lineweight 0.25 mm, Linetype CHAIN).
- Command: PLINE [Enter]
- Draw path passing through Hole 1 horizontally (Y = 40), stepping vertically at X = 75 to Y = 70, passing through Hole 2, stepping vertically at X = 125 to Y = 30, and exiting through Hole 3.
- Make terminating ends and 90? corner vertices thick (0.50 mm) using PEDIT -> Width = 0.5 mm or drawing thick segments.
- Add viewing direction arrows pointing upward (North) and letters 'A' and 'A'.

3. Project the Sectional Elevation (SECTION A-A):
- Switch to layer 'CONSTRUCT' (0.18 mm). Turn on Ortho (F8).
- Project vertical lines downward from all cut hole edges and outer housing walls.
- Switch to layer 'VISIBLE'. Draw the base line, housing height, and outer boundaries.
- Draw all three holes as open visible rectangular passages bounded by thick visible lines (Type A).
- Add vertical chain thin centerlines on layer 'CENTER' at X = 50, X = 100, and X = 150.
- CRITICAL: Do NOT draw any lines at the step locations (X = 75 and X = 125)! The wall appears continuous.

4. Apply Hatching (HATCH command):
- Switch to layer 'HATCH' (Color 8/Cyan, Lineweight 0.25 mm).
- Command: HATCH [Enter] -> Pattern: ANSI31 -> Angle: 0 -> Scale: 1.0.
- Click internal points inside all cut solid wall areas surrounding the three holes.
- Label view: 'SECTION A-A'.

>>**Solution to Question 18:** Discuss the rules for dimensioning sectional views in engineering drawings, detailing how to avoid placing dimension lines across hatch

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- Step 3: Switch to layer 'THIN_BREAK' (Lineweight 0.25 mm, Continuous).
- Step 4: Execute SPLINE (or SKETCH) command to draw an irregular wavy closed or semi-closed boundary enclosing the localized feature.
- Step 5: Command: HATCH [Enter] -> Pattern: ANSI31 -> Pick point inside the area between the feature and the freehand spline break line.
- The local internal feature is now exposed cleanly without altering the rest of the external view.

- Simplified ISO Convention:
* The bearing is drawn as a rectangular envelope matching outer diameter, bore, and width.
* An 'X' or square is drawn inside the envelope representing the rolling element.
* Alternatively, only the inner ring and outer ring cross-sections are hatched, while the balls or rollers are shown unhatched in elevation.

3. Sectional Representation of Helical Springs:
- Helical compression springs are shown with the wire cross-section cut at each coil (drawn as hatched circles or squares).
- For springs with many coils, only the end coils are drawn in section, connected by straight phantom or center lines indicating the spring envelope.

>>Solution to Question 28: Detail the command sequence and drafting steps in AutoCAD to create a Half-Sectional Front View of a stepped flange bushing (Bore 4

[Problem Statement]: Explain the Hatch Boundary Leak detection tools in AutoCAD, discussing Gap Tolerance (HPGAPTOL), red circle glyphs, a
[Detailed Step-by-Step Answer]:
1. Nature of Boundary Leaks:
- When using 'Pick internal point' in the HATCH command, AutoCAD requires a mathematically closed perimeter enclosing the point.
- If two lines fail to meet by even 0.001 mm (an invisible micro-gap caused by manual drafting without OSNAP), the ray-casting algorithm 'leaks' out into the surrounding canvas.
- AutoCAD aborts the operation and displays the error message: 'A closed boundary cannot be determined.'

2. Visual Leak Diagnostics (Red Circles):
- In modern AutoCAD, when a leak occurs, the software automatically places two bright RED CIRCULAR GLYPHS on the screen directly circling the open gap vertices!
- This pinpoints the exact physical location of the boundary defect instantly.

3. Resolution Methods:
- Method 1 (Geometric Fillet Repair):
* Zoom into the red glyph. Command: FILLET -> Radius = 0 -> Select the two open lines. AutoCAD immediately extends or trims them to a perfect closed vertex.
- Method 2 (Gap Tolerance - HPGAPTOL):
* If a drawing contains dozens of tiny drafting gaps:
* Command: HPGAPTOL [Enter] -> Set value (e.g. '1.0' mm).
* AutoCAD will treat any gap equal to or smaller than 1.0 mm as closed, applying the hatch successfully while warning the user.
- Method 3 (Boundary Creation - BO):
* Use the BOUNDARY command to generate a closed polyline boundary before hatching.

>>[Problem Statement]: Explain the methods of creating custom hatch patterns in AutoCAD, detailing the structure of the .pat file, delta-x, delta-y, and line dash
Solution Step Question 25

45, 0, 1.5, 0, 6, 10, 0
- Line 1 draws parallel lines spaced 6 mm apart inclined at 45?.
- Line 2 draws identical lines offset perpendicularly by 1.5 mm, creating the classic double-line steel pattern!

- This occurs because newly created entities are placed at the 'top' of the screen display vector stack.

3. HATCHTOBACK Command:
- Function: Automatically searches the entire drawing database for all hatch objects and moves them to the very BOTTOM of the draw order stack in a single click!
- Command: HATCHTOBACK [Enter]
- All visible outlines, centerlines, dimension lines, and text notes immediately reappear on top of the hatch patterns, completely resolving line obscuration without manual layer shuffling.

4. TEXTTOFRONT Command:
- Accompanying command: TEXTTOFRONT -> Option: 'All' brings all text and dimension annotations to the absolute front of the display stack.

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* Layer properties: Lineweights strictly set to 'ByLayer'.

2. Step-by-Step Assembly Drafting:
- Step 1 (Shaft Centerline Datum): Establish primary horizontal and vertical centerlines on layer 'CENTER'.
- Step 2 (Draw Shaft First): Draw the central drive shaft in external elevation (unsectioned) on layer 'VISIBLE'. Add keyways and shoulders.
- Step 3 (Insert Bearings & Seals):
* Insert standard rolling bearing blocks against shaft shoulders.
* Insert elastomeric oil seals into their housing grooves.
- Step 4 (Draw Housings & End Caps):
* Draw the casting walls of the main gearbox housing and bolted end covers.
- Step 5 (Apply Contrasting Section Lining):
* Main Housing: Hatch with ANSI31 at 45?, scale 1.0.
* Bolted End Cap: Hatch with ANSI31 at 135?, scale 1.0.
* Bearing Rings: Hatch with ANSI32 (steel) at 45?, scale 0.5 (finer pitch).
* Shaft & Fasteners: Left unhatched in elevation.
- Step 6: Execute HATCHTOBACK to ensure all shaft outlines and dimensions remain crisp and prominent.

>> Solution to Question 30: Summarize the complete quality assurance audit checklist for validating sectional views prior to engineering manufacturing release.